Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The value of p and $q(p \neq 0, q \neq 0)$ for which p, q are the roots of the equation $x^2 + px + q = 0$ are

(A) $p = 1, q = 2$

(B) $p = -1, q = 2$

(C) $p = -1, q = -2$

(D) $p = 1, q = -2$

Q.2

If the roots of the equation $\frac{a}{x-a} + \frac{b}{x-b} = 1$ are equal in magnitude and opposite in sign, then

(A) $a - b = 0$

(B) $a + b = 1$

(C) $a - b = 1$

(D) $a + b = 0$

Q.3

Suppose $a, b, c \in \mathbf{R}$ and $b \neq c$. If α, β are roots of $x^2 + ax + b = 0$ and γ, δ are roots of $x^2 + ax + c = 0$, then value of $\frac{(\alpha-\gamma)(\alpha-\delta)}{(\beta-\gamma)(\beta-\delta)}$ is independent of:

(A) a, b

(B) b, c

(C) c, a

(D) a, b, c

Q.4

If $a \in \boldsymbol{R}$ and $a_1, a_2, a_3 \ldots\ldots, a_n \in \boldsymbol{R}$ then $(x - a_1)^2 + (x - a_2)^2 + \ldots + (x - a_n)^2$ assumes its least value at $x =$

(A) $a_1 + a_2 + \ldots\ldots + a_n$

(B) $2(a_1 + a_2 + a_3 + \ldots\ldots + a_n)$

(C) $n(a_1 + a_2 + \ldots\ldots + a_n)$

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(D) none of these

Q.5

Numbers of values of k for which roots of equation $x^2 - 3x + k = 0$ lie in the interval $(0, 1)$ is

(A) only one

(B) no value

(C) finite but more than one

(D) $k \leq \frac{9}{4}$

Q.6

If the roots of the equation $2x^2 - \left(a^3 + 1\right)x + \left(a^2 - 2a\right) = 0$ are of opposite signs, then the set of possible value of a is

(A) $(0, 2)$

(B) $[0, 2]$

(C) $(0, 2]$

(D) $[0, 2)$

Q.7

The value of a for which the equations $x^2 - 3x + a = 0$ and $x^2 + ax - 3 = 0$ have a common root is

(A) 3

(B) 1

(C) -2

(D) 2

Q.8

Ram and Sham solved a quadratic equation. In solving it Ram made a mistake in the constant term only and got the roots as 8 and 2 , while Sham, made a mistake in the coefficient of x only and obtained -9 and -1 as roots. The correct roots of the equation are

(A) $8, -1$

(B) $-9, 2$

(C) $-8, -2$

(D) 9,1

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Q.9

The value of a for which one root of the equation $x^2-(a+1)x+a^2+a-8=0$ exceeds 2 and other is less than 2, are given by

(A) $3<a<10$

(B) $a\geq 10$

(C) $-2<a<3$

(D) $a\leq -2$

Q.10

If x be real and $y=\frac{x^2+6x+1}{2x}$, then which of the following is true

(A) $2<y<4$

(B) $\frac{5}{2}<y<\frac{7}{2}$

(C) either $y\leq 2$ or $y\geq 4$

(D) none of these

Q.11

If a, b, c are all distinct, then $a\frac{(x-b)(x-c)}{(a-b)(a-c)}+b\frac{(x-c)(x-a)}{(b-c)(b-a)}+c\frac{(x-a)(x-b)}{(c-a)(c-b)}-x$ equal to

(A) 0

(B) ax^2+bx+c

(C) $(a+b+c)\left(x^2+x+1\right)$

(D) $\frac{x^2}{a}+\frac{x}{b}+\frac{1}{c}$

Q.12

Number of real solutions of the equation $(12x-1)(6x-1)(4x-1)(3x-1)=5$ are

(A) 0

(B) 2

(C) 1

(D) 4

Q.13

If a, b, and c are not all equal and α and β be the roots of the equation $ax^2+bx+c=0$, then value of $\left(1+\alpha+\alpha^2\right)\left(1+\beta+\beta^2\right)$ is

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(A) 0

(B) positive

(C) negative

(D) non-negative

Q.14

If α, β are the roots of $ax^2 - 2bx + c = 0$, then $\alpha^3\beta^3 + \alpha^2\beta^3 + \alpha^3\beta^2$ is

(A) $\frac{c^2(c+2b)}{a^3}$

(B) $\frac{bc^3}{a^3}$

(C) $\frac{c^2}{a^3}$

(D) none of these

Q.15

The equation $x^{\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}$ has

(A) atleast one real solution

(B) exactly two irrational solutions

(C) exactly one rational solution

(D) complex roots

Q.16

If the equation $x^2 + 2|a|x + 4 = 0$ has integral roots, then minimum value of ' a' is

(A) 4

(B) $-\frac{5}{2}$

(C) 0

(D) -4

Q.17

The number of pairs (x, y) of real numbers that satisfy $2x^2 + y^2 + 2xy - 2y + 2 = 0$ is

(A) 0

(B) 1

(C) 2

(D) none of these

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Q.18

If $b > a$, then the equation $(x-a)(x-b)-1=0$ has

(A) both roots in $[a, b]$

(B) both roots in $(-\infty, a)$

(C) both roots in (b, ∞)

(D) one root in $(-\infty, a)$ and other in (b, ∞).

Q.19

If n is even and $b < 0$, then number of real roots of the equation $x^n + ax + b = 0$ is

(A) 1

(B) 2

(C) 3

(D) none of these

Q.20

The number of integral values of ' a ' for which $(a+2)x^2 + 2(a+1)x + a = 0$ will have both roots integers is

(A) 3

(B) 5

(C) 7

(D) none of these

Q.21

If the equation $\left(x^2+x+3\right)^2 - (\lambda-4)\left(x^2+x+3\right)\left(x^2+x+2\right) + (\lambda-5)\left(x^2+x+2\right)^2 = 0$ has at least one real solution. Then find $[\lambda]$. (where [.] represents greatest integer function)

Q.22

The number of real solutions of $1 + |e^x - 1| = e^x\left(e^x - 2\right)$ is ____.

Q.23

If α is a positive integer and the roots of the equation $6x^2 - 11x + \alpha = 0$ are rational numbers, then the smallest value of α is

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Q.24

For the equation $3x^2 + px + 3 = 0, p > 0$, if one of the root is square of the other, then p is equal to ____.

Q.25

If $\alpha \neq 0$ then the number of values of the pair $\alpha + \beta + \frac{\beta}{\alpha} = \frac{1}{2}(\alpha, \beta)$ such that $(\alpha + \beta)\frac{\beta}{\alpha} = -\frac{1}{2}$, is ____.

Q.26

If the equations $ax^2 + bx + c = 0$ and $x^3 + 3x^2 + 3x + 2 = 0$ have two common roots then $b^2/ac =$ ____.

Q.27

Suppose a and b are real numbers with $ab \neq 0$_If the three quadratic equations $x^2 + ax + 12 = 0, x^2 + bx + 15 = 0, x^2 + (a + b)x + 36 = 0$ have a common negative root then $|a| + |\underline{b}| =$ ____.

Q.28

If a, b, c are positive real numbers, then the number of real roots of the equation $ax^2 + b|x| + c = 0$ is ____.

Q.29

The number of integral values of c for which $\frac{x^2+2x+c}{x^2+4x+3c}$ can take all real values is ____.

Q.30

If α and β are the roots of the equation $x^2 - ax + b = 0$ where $a = 2020$ and $A_n = \alpha^n + \beta^n$, then $(A_{n+1} + bA_{n-1})/A_n$ is ____.

Answers & Solutions

Q.1 Answer:

$p = 1, q = -2$

Solution:

Sum of the roots $= p + q = -p$ **and product of the roots** $pq = q$

$\therefore \quad p = 1 \quad [q \neq 0]$

$\therefore \quad 1 + q = -1 \Rightarrow q = -2 \quad \therefore p = 1, q = -2$

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Q.2 Answer:

$a+b=0$

Solution:

The given equation can be written as $x^2-2(a+b)x+3ab=0$

Since roots are equal in magnitude and opposite in sign.

$\therefore$ **Sum of roots =0**

$\therefore a+b=0$

Q.3 Answer:

a, b, c

Solution:

$x^2+ax+c=(x-\gamma)(x-\delta)$

Thus,

$$\frac{(\alpha-\gamma)(\alpha-\delta)}{(\beta-\gamma)(\beta-\delta)}=\frac{\alpha^2+a\alpha+c}{\beta^2+a\beta+c}$$

$$=\frac{-b+c}{-b+c}=1$$

$[\because \alpha, \beta$ **are roots of** $x^2+ax+b=0]$

Q.4 Answer:

none of these

Solution:

We have

$(x-a_1)^2+(x-a_2)^2+\ldots..+(x-a_n)^2=nx^2-2x\left(a_1+a_2+\ldots.+a_n\right)+\left(a_1^2+a_2^2+\ldots..+a_n^2\right)$

Clearly, $y=nx^2-2x\left(a_1+a_2+\ldots...+a_n\right)+\left(a_1^2+a_2^2+\ldots.....+a_n^2\right)$ **represents a parabola which opens upward. So, it attains its minimum value at the vertex i.e. at**

$x=\frac{2(a_1+a_2+\ldots..+a_n)}{2n}=\frac{a_1+a_2+\ldots..+a_n}{n} \quad \left[\text{using}: x=\frac{-b}{2a}\right]$

Q.5 Answer:

no value

Solution:

As $-\frac{b}{2a}=\frac{3}{2} \notin (0,1)$**, so both roots can not lie between 0 and 1 . So no values of** k **possible.**

Q.6 Answer:

$(0,2)$

Solution:

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The roots of the equation $ax^2 + bx + c = 0$ are of opposite signs if they are real and their product is negative.

i.e. if $b^2 - 4ac \geq 0$ and $\frac{c}{a} < 0$

i.e. if $b^2 - 4ac \geq 0$ and $ac < 0$

i.e. if $ac < 0$

(Q when $ac < 0, b^2 - 4ac$ is automatically ≥ 0)

Hence, the roots of the given equation are of opposite signs if $2\left(a^2 - 2a\right) < 0$

i.e. if $a^2 - 2a < 0$

i.e. $0 < a < 2$.

Q.7 Answer:

2

Solution:

Given equations are $x^2 - 3x + a = 0$ ...(i)

and $x^2 + ax - 3 = 0$ ...(ii)

Subtracting (ii) from (i), we get

$$-3x - ax + a + 3 = 0$$

$$\Rightarrow \quad (a+3)(-x+1) = 0$$

$\Rightarrow$ either $a = -3$ or $x = 1$.

when $a = -3$, the two equations are identical.

So, we taken $= 1$, which is the common roots of the two equations. Substituting $x = 1$ in (i), we get

$$1 + a - 3 = 0$$

$$\Rightarrow \quad a = 2$$

Q.8 Answer:

9,1

Solution:

We know that for a standard quadratic equation $\mathrm{ax}^2 + \mathrm{bx} + \mathrm{c} = 0$ having roots as α and β we have $\alpha + \beta = -\frac{\mathrm{b}}{\mathrm{a}}$ and $\alpha \cdot \beta = \frac{\mathrm{c}}{\mathrm{a}}$

Ramesh did mistake in constant term, but coefficient of x is right, So sum of roots is equal to $8 + 2 = 10$ Sham

And Sham did mistake in coefficient of x, but constant term is right, So product of roots is equal to $-9 \times -1 = 9$

Therefore the actual equation is $\mathrm{x}^2 - 10\mathrm{x} + 9 = 0$ which has roots 9 and 1

Q.9 Answer:

$-2 < a < 3$

Solution:

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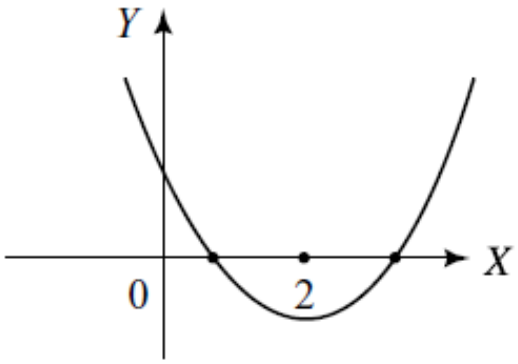
As the roots are real and distinct

$$(a+1)^2 - 4\left(a^2+a-8\right) > 0$$
$$3a^2 + 2a - 33 < 0$$
$$\Rightarrow \quad (3a+11)(a-3) < 0 \Rightarrow -\tfrac{11}{3} < a < 3 \text{ (1)}$$

Also, for

$$2^2 - 2(a+1) + a^2 + a - 8 < 0$$
$$\Rightarrow \quad a^2 - a - 6 < 0 \Rightarrow (a-3)(a+2) < 0$$
$$\Rightarrow \quad -2 < a < 3 \quad (2)$$

From (1) and (2), we get $-2 < a < 3$.



Q.10 Answer:

either $y \leq 2$ or $y \geq 4$

Solution:

$x^2 + x(6-2y) + 1 = 0$ **and** x **is real so,** $D \geq 0$

$$36 + 4y^2 - 24y - 4 \geq 0 \Rightarrow 4y^2 - 24y + 32 \geq 0$$
$$\Rightarrow y^2 - 6y + 8 \geq 0 \Rightarrow y \leq 2 \text{ or } y \geq 4$$

Q.11 Answer:

0

Solution:

Let $f(x) = a\frac{(x-b)(x-c)}{(a-b)(a-c)} + b\frac{(x-c)(x-a)}{(b-c)(b-a)} + c\frac{(x-a)(x-b)}{(c-a)(c-b)} - x$

We have $f(a) = 0, f(b) = 0$ **and** $f(c) = 0$

$f(x)$ **is quadratic equation having more than 2 roots.**

Thus, $f(x) \equiv 0$.

Q.12 Answer:

2

Solution:

$\left(x - \frac{1}{12}\right)\left(x - \frac{1}{6}\right)\left(x - \frac{1}{4}\right)\left(x - \frac{1}{3}\right) = \frac{5}{12.6.4.3}$ **...(i)**

Given the equation Substituting $x = y + \frac{5}{24}$ **in (i), we get**

$$\left(y + \frac{3}{24}\right)\left(y + \frac{1}{24}\right)\left(y - \frac{1}{24}\right)\left(y - \frac{3}{24}\right) = \frac{5}{3.4.6.12}$$

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Or $\left[y^2-\left(\frac{1}{24}\right)^2\right]\left[y^2-\left(\frac{3}{24}\right)^2\right]=\frac{5}{3.4.6.12}$ **...(ii)**

Put $y^2-\frac{5}{24^2}=t$**, then equation number (ii) becomes**

$\left(t+\frac{4}{24^2}\right)\left(t-\frac{4}{24^2}\right)=\frac{5}{3.4.6.12}$

$\Rightarrow \quad t^2-\frac{16}{24^4}=\frac{5}{3.4.6.12} \Rightarrow \quad t=\frac{11}{144}$

$\Rightarrow \quad y^2=\frac{5}{24^2}+\frac{11}{144}=\frac{49}{24^2} \Rightarrow y=\pm\frac{7}{24} \Rightarrow \quad x-\frac{5}{24}=\pm\frac{7}{24}$

$\Rightarrow x=\frac{1}{2},-\frac{1}{12}$

Q.13 Answer:

positive

Solution:

Given data α **&**β **are zeros of polynomial**

ax^2+bx+c

Now sum of zeros $=\alpha+\beta$

$=\frac{-b}{a}$

Product of zeroes $=\alpha\beta=\frac{c}{\alpha}$

Now $\left(1+\alpha+\alpha^2\right)*\left(1+\beta+\beta^2\right)=$

$$=1+\beta+\beta^2+\alpha+\alpha\beta+\alpha\beta^2+\alpha^2+\alpha\beta+\beta^2a^2$$

$$=1+\beta+\alpha+2\alpha\beta+\beta^2+a^2+\alpha\beta^2+\beta^2\alpha^2$$

$$=1+\left(\frac{-b}{a}\right)+(a+\beta)^2-2c\beta\beta+\alpha\beta+\alpha\beta(\alpha+\beta)+(\alpha\beta)^2$$

$$=1-\frac{b}{a}+\frac{b^2}{a^2}-2\frac{c}{a}+\frac{c}{a}+\frac{c}{q}\left(\frac{-b}{a}\right)+\left(\frac{c^2}{a^2}\right)$$

$$=1-\frac{b}{a}-\frac{c}{a}+\frac{b^2}{a^2}+\frac{c^2}{a^2}+\frac{bc}{a^2}$$

$$=\frac{a^2-ab-ac+b^2+c^2-bc}{a^2}$$

$$=\frac{2a^2-2ab-2ac+2b^2+2c^2-2bc}{2a^2}$$

$$=\frac{\left(a^2+b^2-2ab\right)+\left(b^2+c^2-2bc\right)+\left(c^2+a^2-2ac\right)}{a^2}$$

$$=\frac{(a-b)^2+(b-c)^2+(c-a)^2}{\sqrt{2}a^2}>0 \text{ i.e positive}$$

Q.14 Answer:

$\frac{c^2(c+2b)}{a^3}$

Solution:

$ax^2-2bx+c=0$

$\alpha+\beta=-\frac{(-2b)}{a}=\frac{2b}{a}$

$\alpha\beta=\frac{c}{a}$

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$\alpha^3\beta^3 + \alpha^2B^2 + \alpha^3\beta^2$
$\alpha^2\beta^2(\alpha\beta + \alpha + \beta)$
$(\alpha\beta)^2(\alpha\beta + \alpha + \beta)$
$\left(\frac{c}{a}\right)^2\left(\frac{c}{a} + \frac{2b}{a}\right)$
$\frac{c^2}{a^2} \times \left(\frac{(c+2b)}{a}\right)$
$\frac{c^2 \cdot (c+2b)}{a^3}$

Q.15 Answer:

atleast one real solution

Solution:

Let $\log_2 x = t$
then $\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4} = \frac{1}{2}\log_x 2$
$\Rightarrow \quad \frac{3}{4}t^2 + t - \frac{5}{4} = \frac{1}{2t} \Rightarrow 3t^2 + 4t - 5 = \frac{2}{t} \Rightarrow 3t^3 + 4t^2 - 5t - 2 = 0$
A cubic equation has atleast one real solution, $t = 1$ **is solution of the equation other values of** $t = -2, -\frac{1}{3} \Rightarrow x = 2, \frac{1}{2^2}, \frac{1}{2^{1/3}}$ **in which one is irrational.**

Q.16 Answer:

$-\frac{5}{2}$

Solution:

Given equation $x^2 + 2|a|x + 4 = 0$
Since, the condition for integral roots is $b^2 - 4ac = k^2$

$$\Rightarrow (2|\mathrm{a}|)^2 - 4(1)(4) = \mathrm{k}^2$$
$$\Rightarrow 4\mathrm{a}^2 - 16 = \mathrm{k}^2$$
$$\Rightarrow 4\mathrm{a}^2 = \mathrm{k}^2 + 16$$
$$\Rightarrow \mathrm{a}^2 = \frac{\mathrm{k}^2 + 16}{4}$$
$$\Rightarrow \mathrm{a} = \pm\sqrt{\frac{\mathrm{k}^2 + 16}{4}}$$

Take $\mathrm{k}^2 = 9$

$$\Rightarrow \mathrm{a} = \pm\sqrt{\frac{9 + 16}{4}}$$
$$\therefore \mathrm{a} = \pm\frac{5}{2}$$

Q.17 Answer:

1

Solution:

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$x = \frac{-y \pm \sqrt{-(y-2)^2}}{2}$

For x to be real $y = 2 \Rightarrow x = -1$

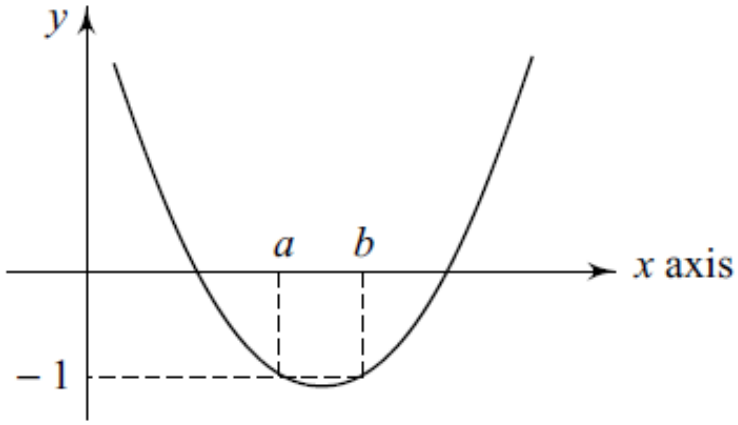
Q.18 Answer:

one root in $(-\infty, a)$ and other in (b, ∞).

Solution:

Graph of $y = (x-a)(x-b) - 1$ is given in Fig.

It is a parabola which open upwards. Also, $y < 0$ for $x = a$ and $x = b$. $\therefore$ $y = (x-a)(x-b) - 1$ meets the x-axis at two points once in $(-\infty, a)$ and once in (b, ∞). Thus, one root lies in $(-\infty, a)$ and one in (b, ∞).



Q.19 Answer:

2

Solution:

Draw graphs $f(x) = \underline{x}^n + ax = 0$ and $g(x) = -b$, we get exactly two points of intersection. Alternatively see that product of roots, which is b, is negative which implies there exists exactly one pair of real roots with opposite signs (since complex conjugates pairs give positive products and if more than one pair of real roots exists then product of roots becomes positive).

Q.20 Answer:

none of these

Solution:

Given equation can be rewritten as $(x+1)\{x(a+2)+a\} = 0$

Since both the roots are integers $\Rightarrow$ $\frac{a}{a+2}$ should be an integer.

$\Rightarrow a = 0, -1, -3, -4$

Q.21 Answer:

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Solution:

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Let $\frac{x^2+x+3}{x^2+x+2} = 1 + \frac{1}{x^2+x+2} = t$

$\Rightarrow x^2 + x + 2 = \left(x + \frac{1}{2}\right)^2 + \frac{7}{4} \geq \frac{7}{4} \Rightarrow t \in \left(1, \frac{11}{7}\right]$

Now equation becomes $t^2 - (\lambda - 4)t + (\lambda - 5) = 0$ ...(i)

at least one root of equation (i) must lie in $\left(1, \frac{11}{7}\right]$

$t = \frac{(\lambda-4) \pm \sqrt{(\lambda-4)^2 - 4\lambda + 20}}{2} = \frac{(\lambda-4) \pm (\lambda-6)}{2} = \lambda - 5, 1$

Hence $1 < \lambda - 5 \leq \frac{11}{7} \Rightarrow 6 < \lambda \leq \frac{46}{7}$

$\therefore \quad [\lambda] = 6$

Q.22 Answer:

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Solution:

$2 + |e^x - 1| = (e^x)^2 - 2e^x + 1 = |e^x - 1|^2$

$\therefore \quad |e^x - 1|^2 - |e^x - 1| - 2 = 0$

or $\quad |e^x - 1| = 2, -1 \quad \Rightarrow |e^x - 1| = 2$

$\Rightarrow \quad e^x - 1 = 2, -2 \quad \Rightarrow e^x = 3, -1 \quad \Rightarrow e^x = 3$

Q.23 Answer:

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Solution:

Since roots of $6x^2 - 11x + \alpha = 0$ are rational and coefficients are rational.

$\Rightarrow 121 - 24\alpha = k^2 \forall k \in I$

Clearly, k is odd, let $k = 2\lambda + 1 \Rightarrow \quad \alpha = 5 - \frac{\lambda(\lambda+1)}{6}$

For minimum value of $\alpha, \lambda = 3 \Rightarrow \alpha_{\min} = 3$

Q.24 Answer:

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Solution:

Let α, α^2 be the roots of $3x^2 + px + 3 = 0$

Now, $S = \alpha + \alpha^2 = -\frac{p}{3}$,

$P = \alpha^3 = 1 \Rightarrow \alpha = 1, \omega, \omega^2$

Now, $\alpha + \alpha^2 = \frac{p}{3}$

$\Rightarrow \omega + \omega^2 = -\frac{p}{3} \Rightarrow -1 = -\frac{p}{3} \Rightarrow p = 3$

Q.25 Answer:

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Solution:

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Take $\mathrm{p} = \alpha + \beta$ and $\mathrm{q} = \alpha\beta$ and check $x^2 - (p+q)x + pq = 0$

$$\Rightarrow x^2 - \frac{x}{2} - \frac{1}{2} = 0 \Rightarrow 2x^2 - x - 1 = 0$$

$$2x^2 - 2x + x - 1 = 0$$

$$2x(x-1) + (x-1) = 0$$

$$x = 1, -\frac{1}{2}$$

Q.26 Answer:

—

Solution:

$x^3 + 3x^2 + 3x + 2 = 0$

$\Rightarrow (x+1)^3 = -1$

$\Rightarrow x + 1 = -1, -\omega, -\omega^2$

$\Rightarrow x = -2, -1-\omega, -1-\omega^2$

As $a, b, c \in \mathbf{R}$, the roots of $ax^2 + bx + c = 0$ are both real or both imaginary.

$\therefore$ roots of $ax^2 + bx + c = 0$ must be $-1-\omega, -1-\omega^2$.

Thus, $-1-\omega-1-\omega^2 = -\frac{b}{a}$

and $(-1-\omega)\left(-1-\omega^2\right) = \frac{c}{a}$

$\Rightarrow \quad 1 = \frac{b}{a}$ and $1 = \frac{c}{a}$

$\Rightarrow \quad a = b = c$

Q.27 Answer:

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Solution:

If α is a common root of the three equations, then $\alpha^2 + a\alpha + 12 = 0, \alpha^2 + b\alpha + 15 = 0$ and $\alpha^2 + (a+b)\alpha + 36 = 0$

$\Rightarrow b\alpha = -24, a\alpha = -21$

Thus, $\alpha^2 - 21 + 12 = 0 \Rightarrow \alpha = \pm 3$

As $\alpha > 0$, we get $\alpha = 3$.

Therefore, $a = -7, b = -8$.

Q.28 Answer:

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Solution:

The given equation $ax^2 + b|x| + c = 0$ can be written as $a|x|^2 + b|x| + c = 0$ [Since $|x|^2 = x^2$] Or, $|x| = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

But $|x|$ can't be $\frac{-b - \sqrt{b^2 - 4ac}}{2a}$ [Since $|x| \geq 0$ for all $x \in \mathbb{R}$].

As it is given to be $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are all positive numbers then $\mathrm{b}^2 - 4\mathrm{ac} < \mathrm{b}^2$ then $-\mathrm{b} \pm$

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$\sqrt{\mathrm{b}^2-4\mathrm{ac}}<0$

So $|x|$ can't be $\frac{-b+\sqrt{b^2-4ac}}{2a}$ also.

So there are no real roots of the given quadratic equation.

Q.29 Answer:

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Solution:

Let y denote the given expression. Then

$y=\frac{x^2+2x+c}{x^2+4x+3c} \quad \Rightarrow \left(x^2+4x+3c\right) y=x^2+2x+c$

$\Rightarrow \quad (y-1)x^2+2(2y-1)x+c(3y-1)=0$

Since x is real, the discriminant of this quadratic is

$4(2y-1)^2-4c(y-1)(3y-1)\geq 0$

$\Rightarrow \quad 4y^2-4y+1-c\left(3y^2-4y+1\right)\geq 0$

$\Rightarrow \quad (4-3c)y^2-4(1-c)y+1-c\geq 0$

which is true for real y if $4-3c>0$, i.e., $\leq \underline{3}^{\frac{4}{3}}$, and

$[4(1-c)]^2-4(4-3c)(1-c)\leq 0$

$\Rightarrow \quad (1-c)[4(1-c)-4+3c]\leq 0$

$\Rightarrow \quad c(c-1)\leq 0 \quad \Rightarrow 0\leq c\leq 1.$

Q.30 Answer:

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Solution:

$A_{n+1}=\alpha^{n+1}+\beta^{n+1}$

$aA_n-bA_{n-1}=a\left(\alpha^n+\beta^n\right)-b\left(\alpha^{n-1}+\beta^{n-1}\right)$

Now

$\alpha+\beta=\mathrm{a}, \alpha\beta=b \quad \therefore \quad aA_n-bA_{n-1}=(\alpha+\beta)\left(\alpha^n+\beta^n\right)-\alpha\beta\left(\alpha^{n-1}+\beta^{n-1}\right)$

$=\alpha^{n-1}+\alpha\beta^n+\alpha^n\beta+\beta^{n+1}-\alpha^n\beta-\alpha\beta^n=\alpha^{n+1}+\beta^{n+1}=A_{n+1}$

$\therefore \quad A_{n+1}=aA_n-bA_{n-1}$

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